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NPTEL

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REINFORCEMENT LEARNING

Options

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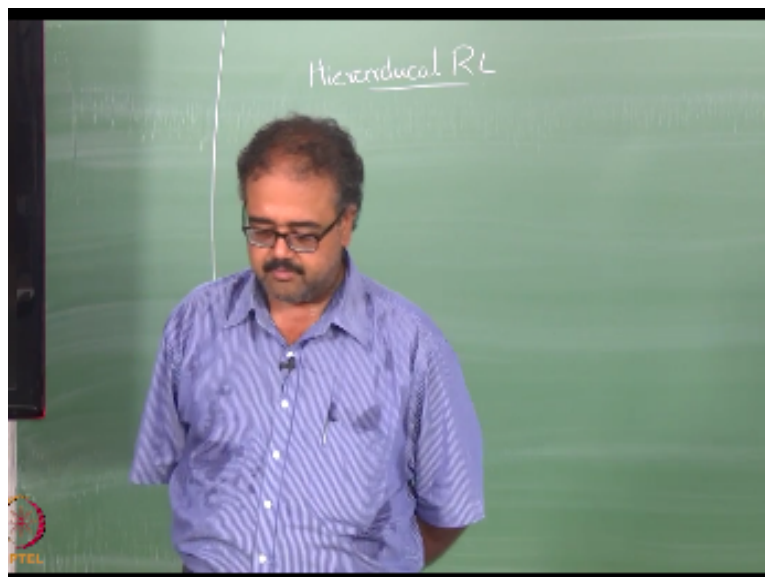
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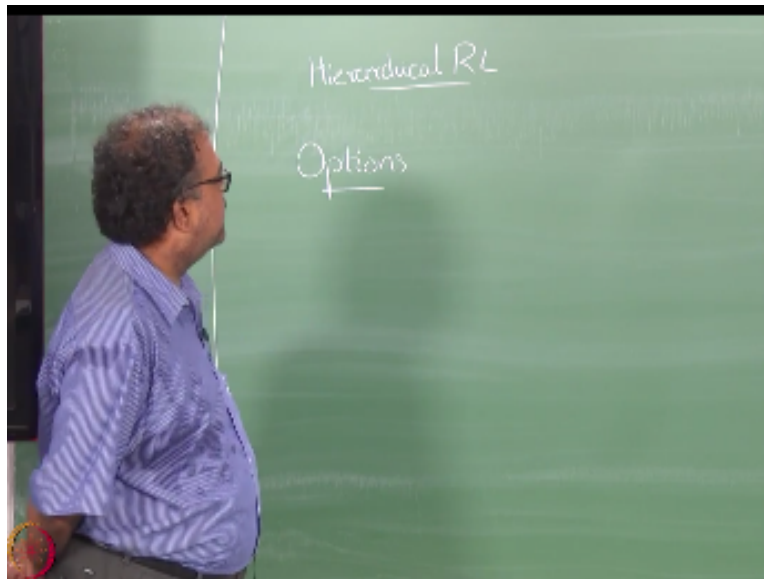
So we are going to continue looking at hierarchical reinforcement learning today and so.

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What we saw in the last class was essentially a very long drawn motivation for why you want to do hierarchies right and then I spoke about a lot of issues that might creep up right so it is more at a very, intuitive level right. And finally towards the end of the lecture we actually looked at SMDPs right so, which is a model that many of the hierarchical learning architectures use as the underlying, mathematical formalism okay.

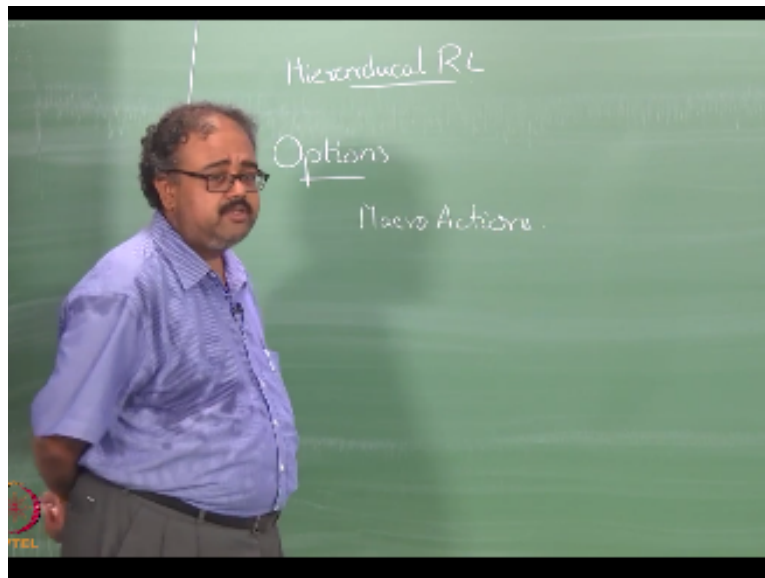
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So what we will do today is start looking at some of those some of those hierarchical frameworks okay the first one we will look at is its called the options framework. Chronologically this is not the earliest but in some sense it is the simplest of the hierarchical frameworks. And therefore we will start there so the basic idea behind options is something very, very, very simple.

So I am going to take the solutions to these sub problems that we are defining right so I just take the solutions to those right I will encapsulate them as a single action so the idea comes from the planning literature actually where people talk about something called macro actions.

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Right so in the planning community macro actions are essentially just a sequence of actions put together and then we talk about it as a single action right so for example we spoke about many such examples in the last class right and so we want to get down the go out of this building I can have a macro action that says climb down the stairs.

Right that consists of a lot of smaller actions but then as far as I am concerned all I need to think of it is just a single action that says climb down the stairs right so and in the planning literature you just have a sequence of actions right but then when we move to the reinforcement learning space there is something else that we need to take care of what is that I can't just blindly specify a sequence of actions let us take the go down the stair case.

Can I just say okay standing front of the stand at the top of the stairs okay then put your right foot out down once put your left foot down once and keep doing this. I don't know..120 times and you will reach the ground floor right. So can I specify a macro action like that, what could go wrong? Whatever okay let's assume that I get the floor right, is there anything else that can go wrong, balance let's assume I get the balance right.. sir direction, lets assume you get the direction right but

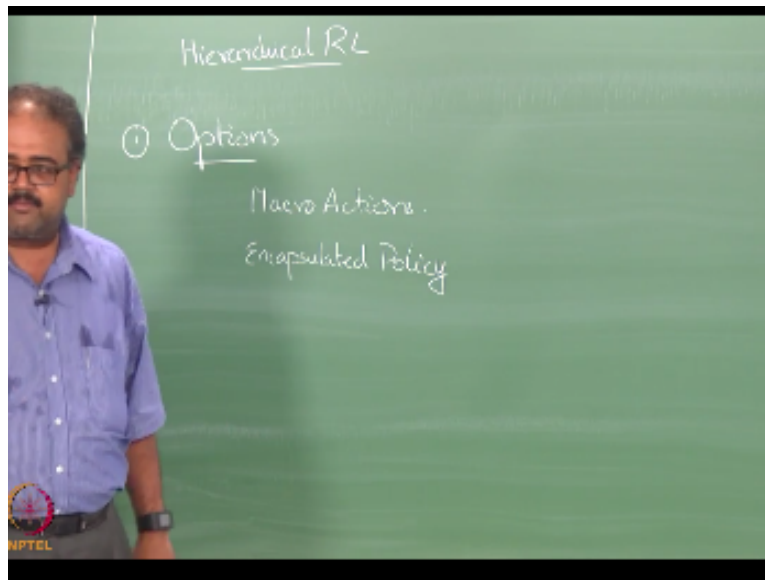
When can the direction go wrong.. yeah I'm assuming I've learnt all of that people could just becoming up the stairs right I might have to stop move to the side let the more I can't just keep bulldozing.. no these are things that are dynamic right see the direction the stair turns and all those things are static so I could I could plan for all of that and then say okay.

This is the direction in which I will be taking my steps and so on so forth right so all of that I could plan for but then there are other dynamic components to the world right and suppose there is some kind of slipperiness associated with the stairs right so I keep going down and then suddenly some stair I could skid and then I might actually fall down a few steps I mean not really fall down but then like skip ahead a few steps right or suddenly there could be I don't know this happens.

In the other day it happened to me when I was climbing down the civil engineering department suddenly there is this one, one step there where this is big chunk of concrete taken out of the step right. I was talking to somebody and then I was actually holding a cup of coffee in my hand and walking suddenly there is no step there okay I skip the step and actually had to step on the next, next step this kind of puts your stride off right and then I had to actually go down a few more steps very quickly to regain my balance so these kinds of things happen right

These are all dynamic things you do not know small changes that are happening and so on so forth we are actually all this while when we are talking about solving the reinforcement learning problem we have been assuming we live in a stochastic world right so, so things can go wrong right so what do you need to specify to get a macro action right, not a sequence of actions but start and then the end I need the solution to the problem right close observations right.

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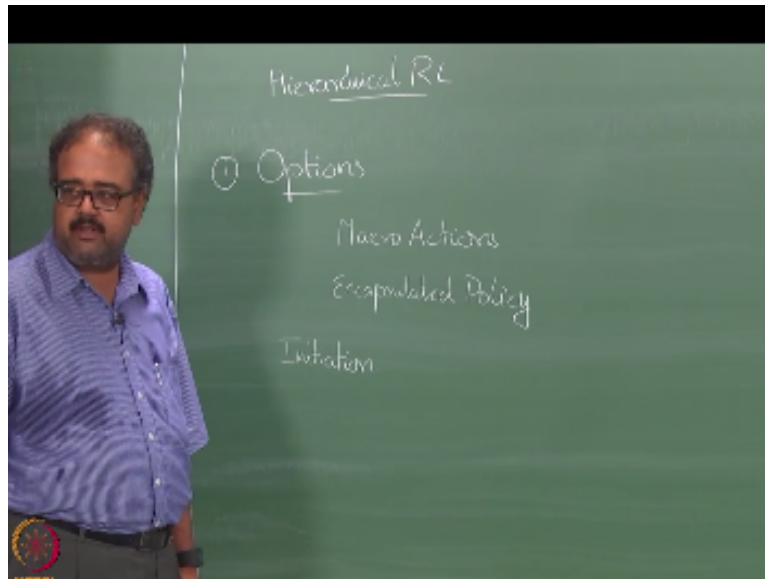


So what you need to specify is a policy right so an option is essentially some kind of a I'm going to call it some kind of an encapsulated policy okay great so now let's go back to the example I was actually giving you already so how did I describe this macro action for getting downstairs. How did I start the description I say stand in front of the step so I need to be able to tell you where these macro actions are applicable right I need to tell you where they are applicable I can't just say ok get down the stairs now right so if I say get down the stairs here.

I mean it really it won't work right so I have to be actually there this is not a new, new idea even for simple actions I need to be able to specify where the action is valid right so I cannot just say when playing chess and the first move cannot be capture the king right so, so I have to set it up and there has to be a mate condition before I can say go cut the king right so, so those kinds of things I need to tell you where to start and then I need to tell you where to end right.

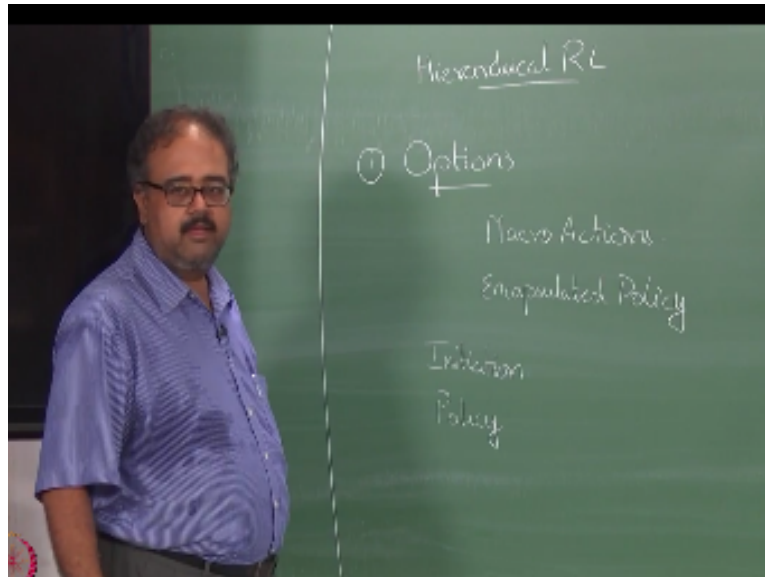
If I if I, exactly what I was telling you if I do this 130 times and then you will end at the ground floor right so I need to tell you where to end so I put all of these things together so if I want to specify a useful macro action.

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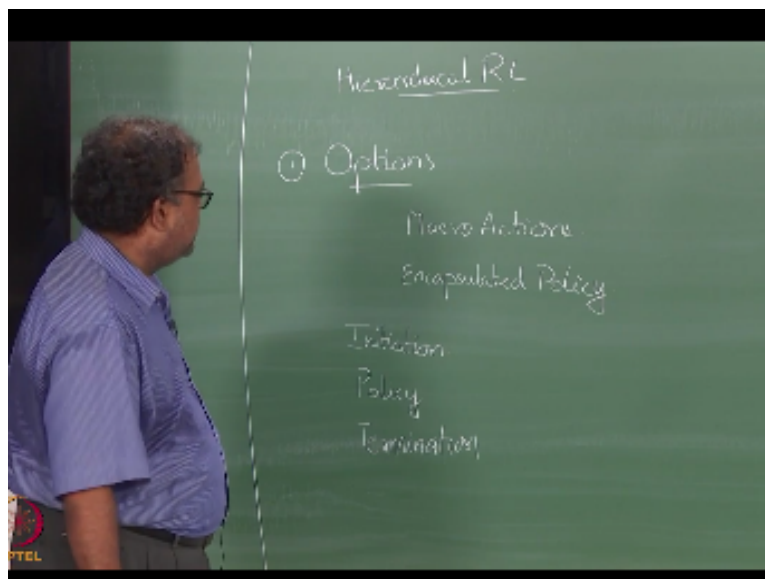
So I need something that tells me where the initiation can happen right then

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I need a policy to use while the macro action is executing and then

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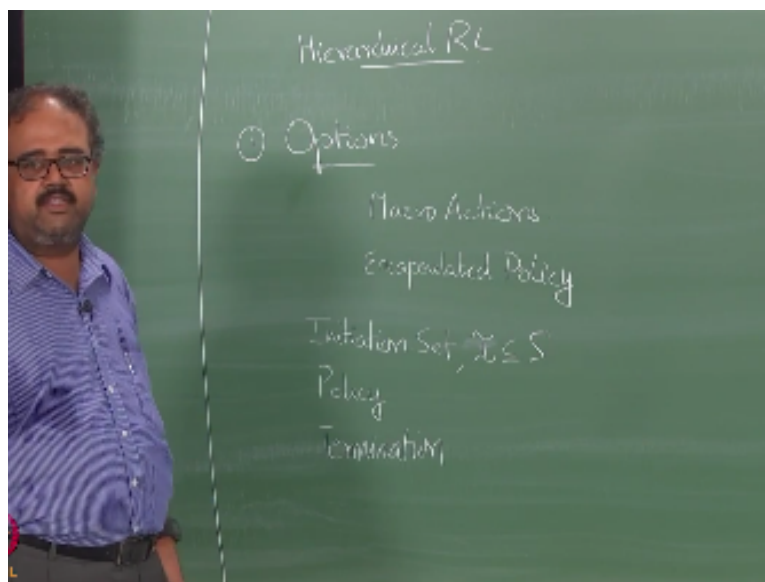


I need I need a termination ok sorry it can be started in states I agree but why what do you mean we within the particular encapsulation, the encapsulation is just a policy right just a policy so

assume I assume that wherever I can start the policy I will have some action defined for that policy right so let's talk about that little further as we go along right so,

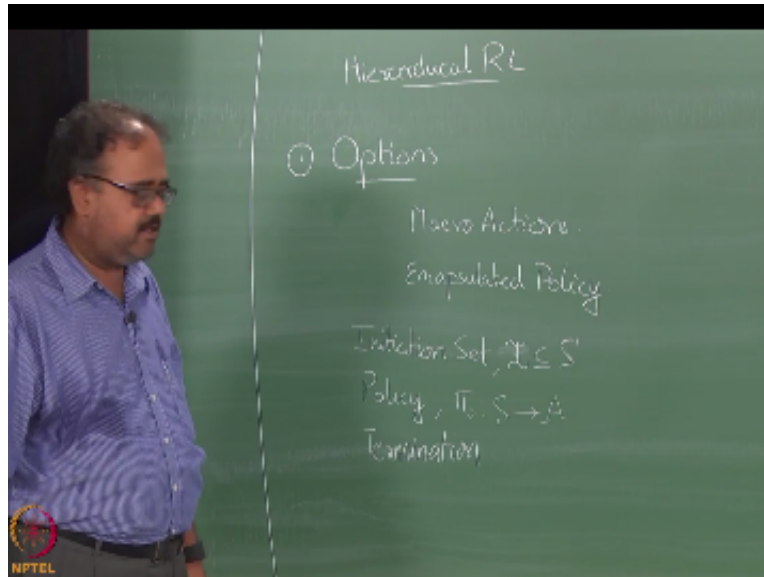
So the initiation if you look at the, the classical options framework right I mean it is proposed in '99 proposed I suppose we can start calling it classical now if you look at the classical options framework they usually give this as

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A set of states right so initiation set is denoted by I is a subset of the state space okay. $I_0 \subset S$

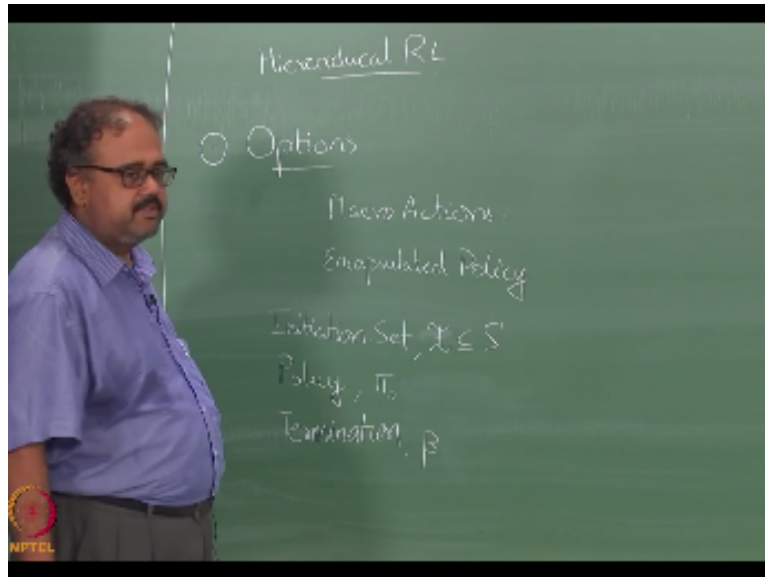
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And then the policy is given as, if it is a deterministic policy it could be this right or it could also be given as a mapping of yeah or it also be given as a mapping of history I will elaborate a little bit more, it can be given as a mapping of a history of things that have happened since I started doing this to the next action right so that will allow us to define options like you go three steps forward and then, turn right.

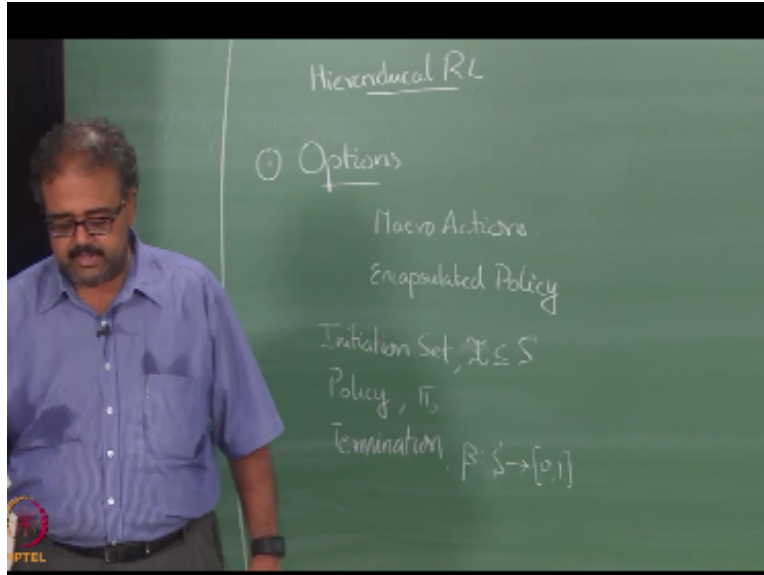
It might be useful right so what happened there when I said go three steps forward and turn right automatically added history to the policy so I need to count okay how what was my last action was it go forward okay what was the last two actions were they are both go forward what are the last three actions were they both they all three go forward right and then if that is the case then I do a right action right so I could define it like that I will come back to that and

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The termination is typically specified by β right $\beta_o : S \rightarrow [0, 1]$

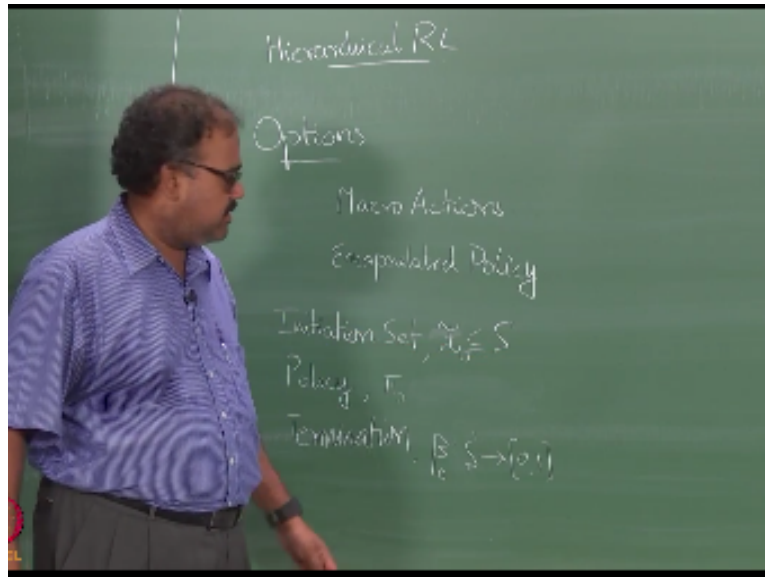
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Is in the original definition of option is given by a probability. So $\beta(s)$ will be the probability that that option will stop in state s right so for many states the probability could be one because that means on all those states the option if you reach that state the option will not continue. Yeah but this is not the RL problem I am solving right, no no..it is just an action I am defining it is not a problem so suppose you say.

You cannot tell me that Oh in the agent if it goes north you cannot you should not say that its actually move up one step right I mean that is the dynamics of the problem I am just defining the dynamics here so right so these are the three components that make up an option right so for a particular option.

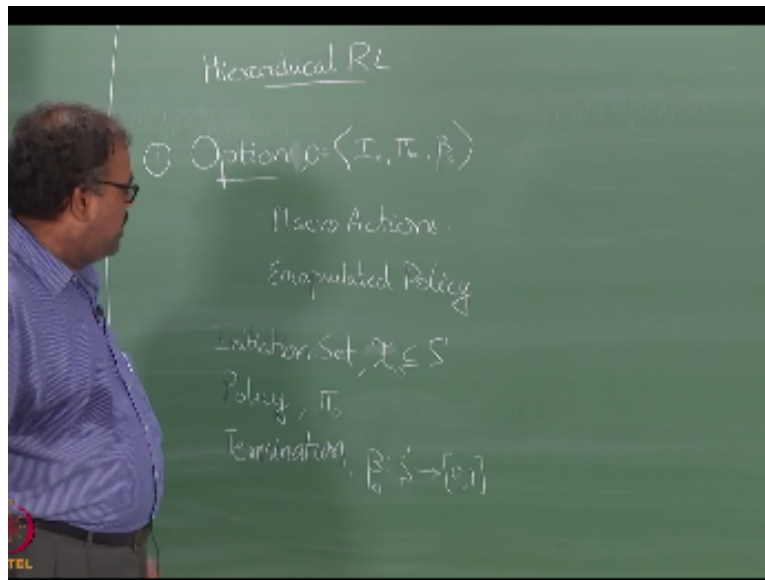
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I can probably add a o to everything so I have the basic MDP with some states actions and so on so forth and an option is going to consist of

$$O = \langle I_o, \pi_o, \beta_o \rangle$$

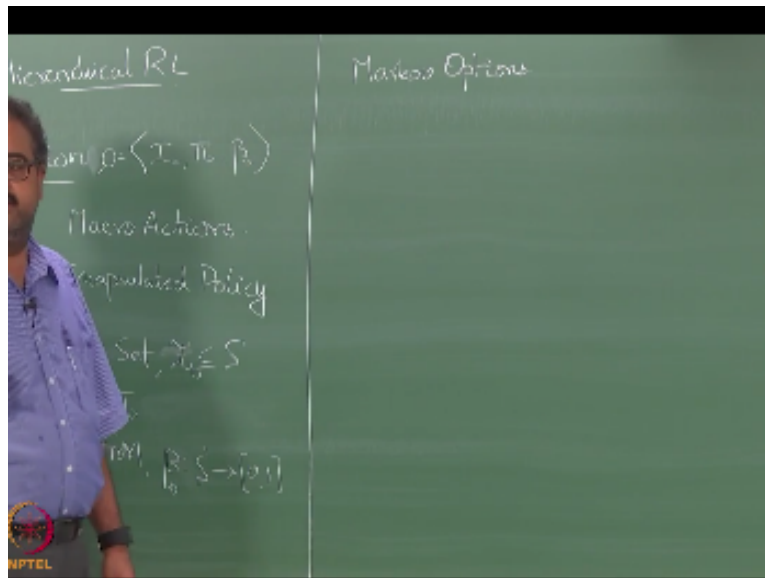
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These three things so this probabilistic definition of options was actually introduced to for more of a mathematical convenience right so, so that they could show some proofs more elegantly in practice. When you define this options you typically make the probability of termination 0 or 1 so 0 means it does not terminates, 1 means it terminates right so you can think of it as an indicator function that indicates whether the, the set belongs to the termination set or not.

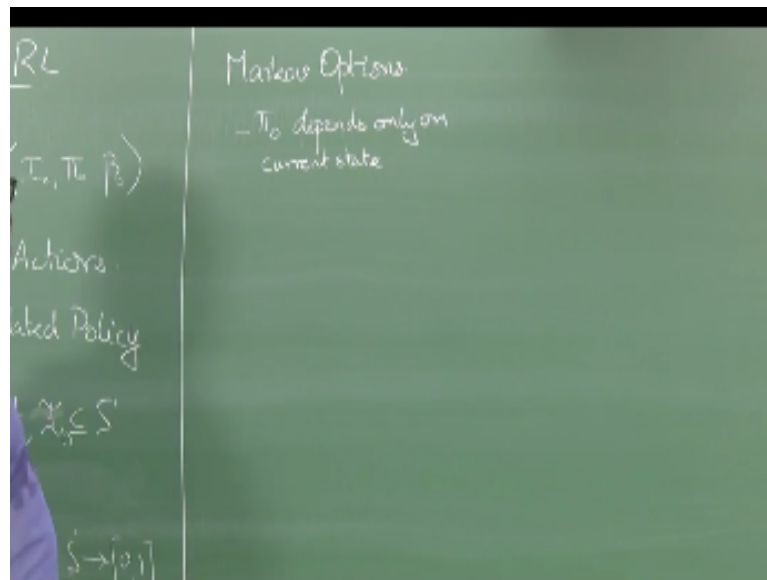
Right so you could think of the initiation and the termination to be sets okay.

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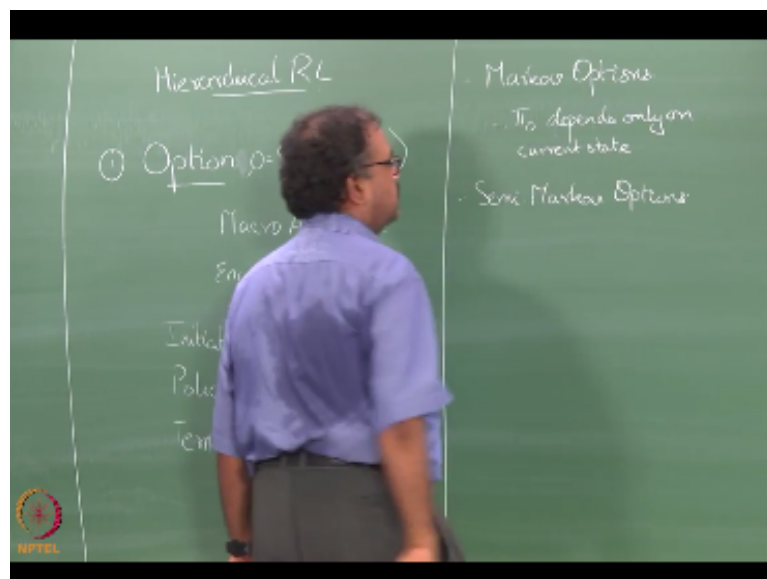


So there are two kinds of options right so which I will, markov options which are easy enough to define so what are markov options. Sorry, exactly so π_0 is defined only on the current state right so these are markov options so the nice thing about Markov options is I do not have to worry about where I started the option right so wherever it is it will be the same action right so if I am in a particular state and I am executing that option then I will take the same action.

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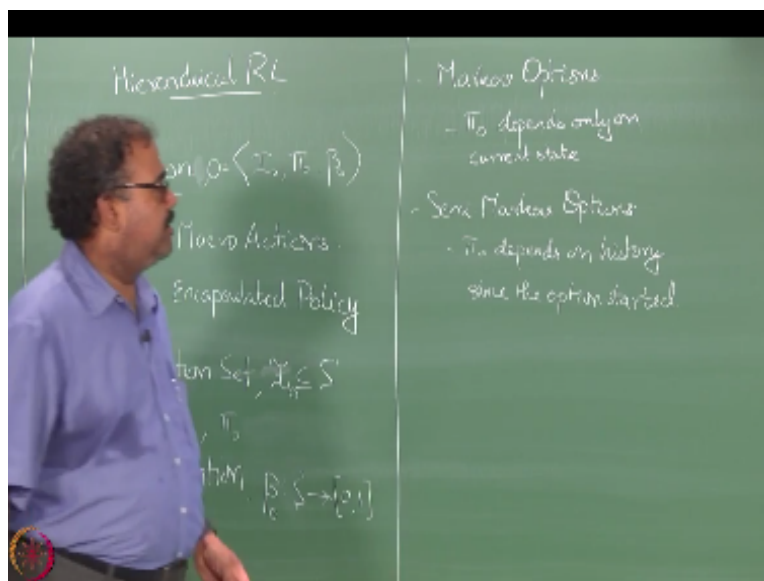


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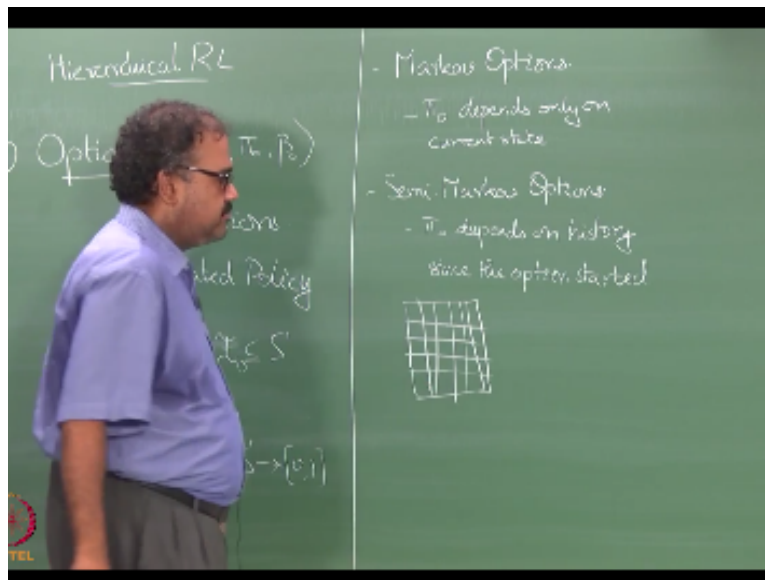
In the second class of options called semi Markov options. So what are semi markov options, why semi markov why not non markov, sequence of states since the option was initiated right if you remember I was telling you why it is a semi Markov because during the execution of the option there is a dependence on history or could be a dependence on history we do not know right I was telling you at the I mean when the holding time is an operation you really do not know how the system is behaving.

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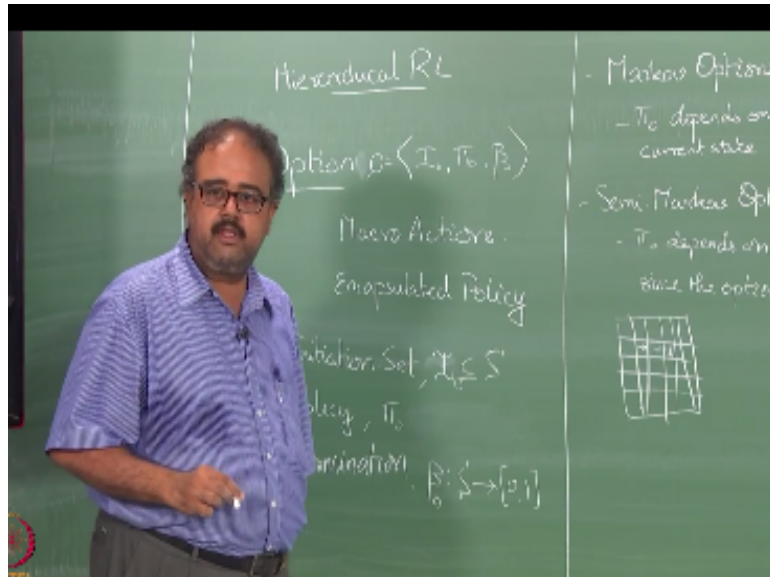
So and I say that is the reason we are calling it semi Markov so in a semi Markov option so π_0 depends on history since the option started so let us say

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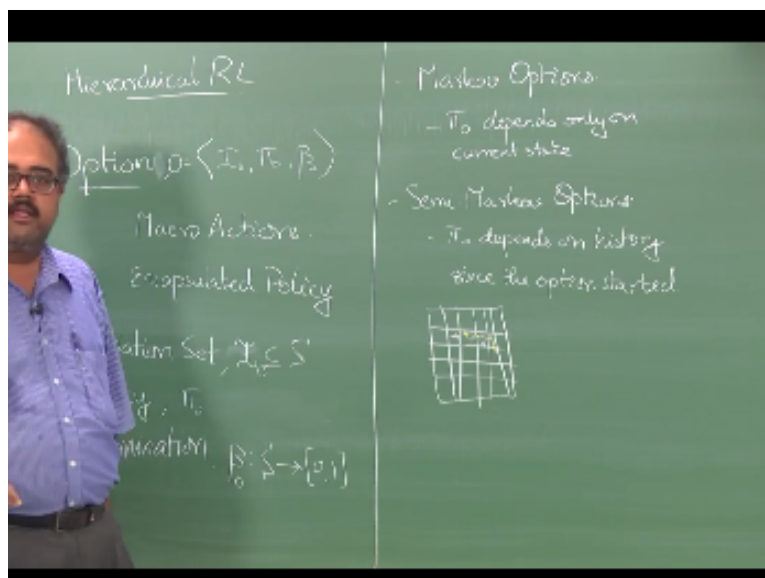
I have a if you start here and say go right two steps and down one step right

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So it will be this, this like 1 2 and then this right so in this state I go right in this state I go down right but if I start at the option.

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Here and said go right two steps and down one step right it will be this, this and then that right so what I do in this state depends on where I started the option right. Whether I go right here or whether I go down here depends on where I started the option if it had been a markov option right it should not matter where I start the option when I come to this state I should be taking the same action.

So there are many instances where we want to allow your agent just to do a larger amount of exploration okay so how many of you have started finished all your programming home works Q learning right if you make me think of how hard it would be if you make the puddle world 10,000 x 10000 puddle world did people try it isnt this part of the assignment what is part of the assignment then well we didn't give them a function approximation assignment till now, on the 12 by 12.

What's the fun in that when did you change the problem statement I thought I thought I gave a 1000 x 1000 at least a thousand by thousand 12 x 12 there is no fun why do you even want to do function approximation on a 12 x 12 it for atleast a 1000 x 1000 grid world. What is it that's the fun, that it is the fun part what do you mean by store the policy why do they need to store the policy they just need the Q function right.

That is book keeping, I do not care, I mean you just ask them to plot learning curves right. That should be good enough for the function approximation case, we should not ask them to return the policy you are very smart computer science kids cant you figure out how to do that automatically right I mean just look at it blow up the puddle and figure out rules for where the puddle should be right and then write a small function that initiates the puddle world that is easy enough to do do not tell me you cannot do it right anyway how long would Q-learning take to work in the 1000 by 1000 grid world or a 10000 by 10000 grid world.

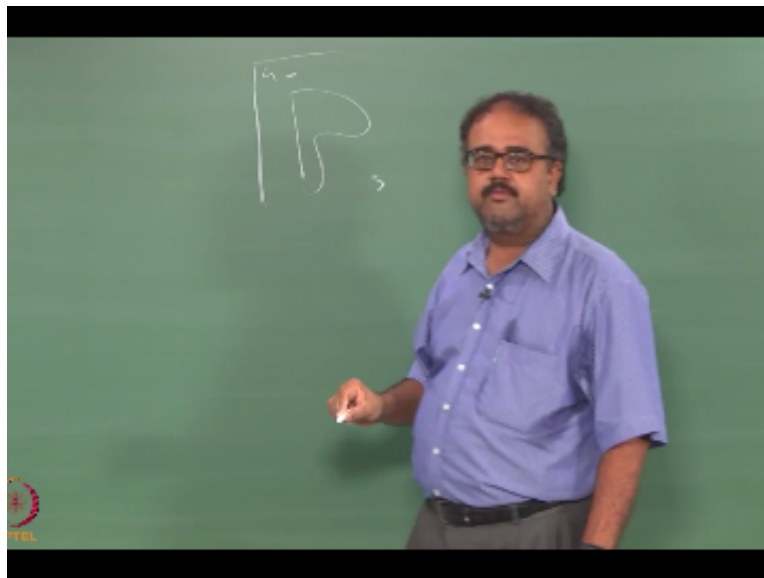
We saw how long it takes right it takes even in a 12 by 12 grid world it just takes awhile to do all the exploration and so on so forth in situations like that you really want your agent to have the ability to do crazy things in one shot right so for example imagine a 1000 by 1000 grid world

where I define an option that says go in this specific direction for fifteen steps so I can just go north 15 steps or go east 15 steps.

See in the puddle world is not really clear to me what are useful options to begin with right I mean what are the useful options, there are no doorways what are useful options in the puddle world now if the puddle repeats but the kind of a puddle world I gave in the RL exam that one big puddle in the middle of the world really no bottlenecks or anything that.

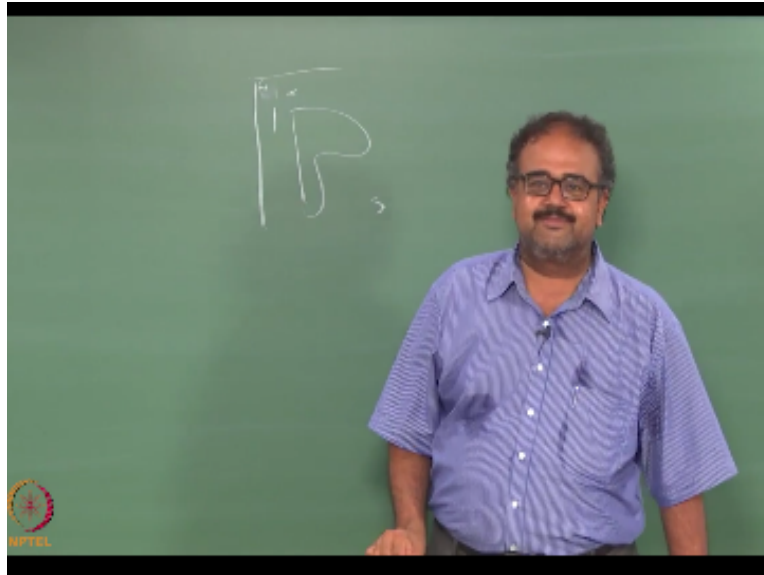
I mean there are bottlenecks but it is not immediately obvious to you what the bottlenecks are the bottlenecks would be the, the edge of the puddle before it ends.

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So essentially you have the puddle like this right, right and then there is a goal here and the start state here right this would actually be a bottleneck if you think about it. Right so this could be a bottleneck so you need to get there before you can actually get to the goal okay.

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So that will be that will act that those states will act like bottlenecks right so but then it is not that is not useful really that interesting an option right so what you could do is just define or the goal was here is it wherever you know you kind of can guess where the option bottlenecks would be right so you can kind of define options that essentially make it more efficient to explore by saying that go up 15 steps right or go right 15 steps right.

So that way instead whenever you pick this action by accident when do you pick it by accident when you are doing exploration right your epsilon greedy thing when you pick go up the 15 state suddenly you break out of your random walk loop right so you move away 15 steps and then you start doing a random walk there right and then you say you take another action then you will move away another 15 steps and you start doing a random walk there that way you essentially

can explore lot more of the state space then you can if you are just working with primitive actions so in such cases having these kinds of options are helpful.

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